Homomorphic Sensing: Extension to Union of Subspaces Case, Sparse Signal Recovery and Probabilistic Analysis

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Abstract

Homomorphic sensing (HS) studies conditions which guarantee unique recovery of a signal from its homomorphic image under an unknown transformation in a finite set. It finds applications in many problems which arise in data science/engineering such as record linkage and image registration. In this thesis we contribute to HS in two aspects. Firstly, we extend HS to the case where the input space is a *union* of linear subspaces. One of the applications of this work is in uniqueness guarantees for a class of sparse signal recovery problems. Secondly, we study HS under noise, and develop both deterministic and probabilistic results which characterize the relation of the signal-to-noise ratio and the estimation error of a natural estimator of the original signal.

1 Introduction

1.1 The Problem of Homomorphic Sensing

Homomorphic sensing (HS) [17] studies at an abstract level uniqueness guarantees for recovering signals from their homomorphic images. To formally introduce what HS is concerned with, we first have the following definition.

Definition 1. Let $\mathcal{V} \subset \mathbb{C}^m$ be a set of vectors in $\mathbb{C}^m$ and $\mathcal{T}$ be a finite set of linear transformations of $\mathbb{C}^m$. We say that "v_1 is uniquely recoverable in $\mathcal{V}$ under $\mathcal{T}$ " if whenever $\tau_1(v_1) = \tau_2(v_2)$ for some $\tau_1, \tau_2 \in \mathcal{T}$ and $v_2 \in \mathcal{V}$, then $v_1 = v_2$.

Intuitively, a signal v is uniquely recoverable if it can be uniquely determined by its homomorphic image that we observe. HS, then, studies the conditions which guarantee that all, or almost all vectors in $\mathcal{V}$ are uniquely recoverable. By specializing $\mathcal{V}$ and $\mathcal{T}$, HS finds applications in many problems which arise in data science/engineering, which we introduce in what's next.

Unlabelled sensing [19] is a specialization of HS when $\mathcal{T}$ is the set of selection matrices[1] and $\mathcal{V}$ is a linear subspace. It studies the recovery of a signal from its permuted and sub-sampled linear measurements. When there is no sub-sampling, $\mathcal{T}$ becomes the set of permutations and the problem becomes *linear regression without correspondence*, or *shuffled linear regression* [4, 12, 1, 18]. It arises in applications such as image registration [8], pose/correspondence estimation [2, 10], multi-target tracking [13] and record linkage for data integration [6, 15].

Real phase retrieval [9] aims to recover a signal $x^* \in \mathbb{R}^n$ up to sign from the magnitudes $y \in \mathbb{R}^m$ of its linear measurements. It can also be formulated as a specialization of HS where $\mathcal{T} = \mathcal{B}$ is the set of sign matrices[2] and $\mathcal{V}$ is a linear subspace. *Symmetric mixture of two linear regressions* [5] also has this formulation. Formally, we observe

$$y = B^* A x^* \tag{1}$$

[1] A selection matrix is equivalent to a permutation matrix composed by a coordinate projection.

[2] A sign matrix is a diagonal matrix with 1 or -1 diagonal entries.

where $B^* \in \mathcal{B}$ is an unknown sign matrix, $A \in \mathbb{R}^{m\times n}$ is a known basis matrix for the linear subspace $\mathcal{V}$, and we want to recover (up to sign) $x^* \in \mathbb{R}^n$, the unknown original signal.

In the above examples, $\mathcal{V}$ is a linear subspace, which is not a tight description in some situations. For example, when we want to consider the recovery of *sparse* signals, $\mathcal{V}$ is in fact a *union* of linear subspaces. Let us illustrate with further details. Suppose x^* is k-sparse[3], and we observe

$$y = TAx^* \tag{2}$$

where T is a unknown transformation of $\mathbb{C}^m$ in $\mathcal{T}$ and A is a known $m \times n$ matrix with full colomn rank. Here since x^* is k-sparse, Ax^* is inside the subspace spanned by the k columns of A indexed by the k non-zero entries of x^*. Since we don't know which k entries of x^* are non-zero, there are $\binom{n}{k}$ possible k-dimensional subspaces for Ax^* to live in, and therefore Ax^* is inside the union of these $\binom{n}{k}$ linear subspaces. Note that although one can still say that Ax^* is inside the subspace spanned by all columns of A (i.e., the range space of A), this is not a tight description as the sparseness of x^* is not taken into account. By replacing $\mathcal{V}$ in previous examples (as a linear subspace) by this union of subspaces as we described, previous specializations become recoveries of sparse signals in the corresponding settings, for example, real phase retrieval for sparse signals which is studied in [7, 11].

All above discussions assume noiseless model which is inherently impractical in real cases. There are plenty of studies of noisy models in some specializations of HS. A common modelling assumption is that a noise vector ϵ is added to the signal that we observe, so we have $\bar{y} = T^*Ax^* + \epsilon$ and want to recover x^* from $\bar{y}$. A natural estimator of x^* and T^* is the maximum likelihood estimator:

$$(\hat{T}, \hat{x}) \in \underset{T\in\mathcal{T}, x\in\mathbb{C}^n}{\arg\min} ||\bar{y} - TAx||_2 \tag{3}$$

In unlabelled sensing, [19] shows local stability of this estimator, specifically, $||\hat{x} - x^*||_2$ goes to zero when $||\epsilon||_2$ goes to zero. In the context of linear regression without correspondence, [12] studies the snr requirement for recovering the correct permutation matrix which is different but related to the focus of HS. Also aiming at permutation recovery, [16] shows that when the permutations are sparse, i.e., they permute only a small percentage of the entries of Ax^*, the snr requirement could be less strict. We note that currently there is no study of HS under noise at a general level to our knowledge.

1.2 Contributions and Outline

[17] first proposes the study of HS and gives theoretical conditions for uniqueness guarantees for the case where $\mathcal{V}$ is a linear subspace. Based on this work, we contribute to HS in two aspects.

- Firstly, in Section 2, we extend HS to the case where the input space is a *union* of linear subspaces. One of the applications of this work is uniqueness guarantees for a class of sparse signal recovery problems as we briefly discussed earlier.
- Secondly, we study HS under noise in Section 3. We first develop deterministic conditions which guarantee that the difference of $\hat{x}$ and x^* stays "small" in a sense, and then perform probabilistic analysis under the assumption that the transformations (elements in $\mathcal{T}$) has only 1/0 singular values which applies to existing instances of HS. Specifically, we show how to reduce the analysis to the case of $n = 1$, and develop probabilistic results which characterize the relation of the signal-to-noise ratio and the estimation error when the transformations are additionally full rank. We also discuss about the case where the transformations may not have full rank, and expect that a result similar in spirit could be obtained, which we put as future work.

1.3 Preliminaries and Notations

Most terminologies we use for technical development largely follows [17], where for linear algebra terms they can also be found in [14]. We introduce part of them which will be used frequently. A linear map from a vector space to itself is called an endomorphism, and an invertible endomorphism is called an automorphism. For an endomorphism τ of $\mathbb{C}^m$, its kernel is the set of vectors which maps to 0, i.e., $\ker(\tau) = \{v \in \mathbb{C}^m : \tau(v) = 0\}$, and its image is $\mathrm{im}(\tau) = \{\tau(v) : v \in \mathbb{C}^m\}$. Then rank

[3] a vector is k-sparse if it has at most k non-zero entries.

of an endomorphism is the dimension of its image $\mathrm{rank}(\tau) = \dim(\mathrm{im}(\tau))$. The eigenspace of τ associated with eigenvalue λ is the vector space $\mathscr{E}_{\tau,\lambda} = \{v \in \mathbb{C}^m : \tau(v) = \lambda v\}$. The codimension of a subspace $\mathcal{V} \subset \mathbb{C}^m$ is the ambient dimension minus its own dimension: $\mathrm{codim}(\mathcal{V}) = m - \dim(\mathcal{V})$. For a matrix A, we use P_A for the orthogonal projection onto $\mathcal{R}(A)$, and use $P_{A^\perp}$ for the orthogonal projection onto $\mathcal{R}(A)^\perp$. For some n dimensional subspace $\mathcal{U}$ in $\mathbb{C}^m$, we call $A \in \mathbb{C}^{m\times n}$ a basis matrix for $\mathcal{U}$ if the column vectors of A span $\mathcal{U}$. The singular value decomposition (SVD) if a matrix is denoted by $U_X \Sigma_X V_X^T$, where Σ_X is a diagonal matrix and its diagonal entries are the singular values of X. Sometimes we also use $U_X \Sigma_X V_X^T$ for the thin-SVD of X (will be noted), this form discards 0 singular values and associated left and right singular vectors in U_X and V_X, therefore $U_X, V_X \in \mathbb{C}^{m\times r}$ and $\Sigma_X \in \mathbb{C}^{r\times r}$ where $r = \mathrm{rank}(X)$. The column space of X is $\mathcal{R}(X)$ and we also sometimes use $\mathrm{Span}(X)$. The null space of X is denoted as $\mathcal{N}(X)$. The standard inner product of matrices is denoted by $< X, Y >= \sum_{i,j} X_{ij} Y_{ij}$.

We will also use tools from Algebraic Geometry [3], where a very important fact is that the intersection of finitely many Zariski non-empty open sets is a Zariski non-empty open set. An algebraic variety of $\mathbb{C}^m$ means the zero locus of a set of polynomials in m variables. $\mathrm{Gr}(m, n)$ refers to the set of n-dimensional linear subspaces of $\mathbb{C}^m$, which is an algebraic variety and is called Grassmannian.

A finite union of linear subspaces, say $\mathcal{V} = \bigcup_{i=1,\dots,l} \mathcal{V}_i$ as we will frequently mention in Section 2, is generally not a linear subspace but is an algebraic variety. The dimension of $\mathcal{V}$ refers to its algebraic dimension, which is in fact the largest dimension of $\mathcal{V}_i$, so formally, $\dim(\mathcal{V}) = \max_{i=1,\dots,l} \dim(\mathcal{V}_i)$. A point is generic in a space if it is inside a set with measure 1, so one can think of "generic" as "randomly drawn from a continuous probability measure". In this paper since we are in the context of algebraic geometry, this measure-1 set will usually be a non-empty open set in the corresponding space in the Zariski topology. For example, when we say a generic n-dimensional linear subspace $\mathcal{V} \subset \mathbb{C}^m$, it means that $\mathcal{V}$ is inside a non-empty open subset $\mathbb{U} \subset \mathrm{Gr}(m, n)$ in the Zariski topology of $\mathrm{Gr}(m, n)$. When we say a generic point v inside a union of linear subspaces $\mathcal{V} = \bigcup_{i=1,\dots,l} \mathcal{V}_i$, we mean that there exists non-empty open sets $\mathbb{U}_i \subset \mathcal{V}_i$ where $v \in \bigcup_{i=1,\dots,l} \mathbb{U}_i$. We will also frequently use the following conclusion.

Lemma 1. *Let $\mathcal{Y}$ be a variety defined by homogeneous polynomials and $\mathcal{V}$ a generic linear subspace. Then* $\dim(\mathcal{Y} \cap \mathcal{V}) = \max\{\dim(\mathcal{Y}) - \mathrm{codim}(\mathcal{V}), 0\}$.

Finally, we use C to represent universal constants which could change from line to line.

2 Homomorphic Sensing of a Finite Union of Linear Subspaces

2.1 Sensing under Automorphisms

We first consider the conditions for unique recovery in the base case where $\mathcal{T}$ consists of the identity mapping and an automorphism and also $\mathcal{V}$ is the union of two subspaces, which is easy to build on top of existing results.

Lemma 2 (Extension of Lemma 2 in [17])**.** *Let τ be any automorphism of $\mathbb{C}^m$ and let $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$ be a union of two linear subspaces $\mathcal{V}_1$ and $\mathcal{V}_2$, $\mathcal{V}_1 \neq \mathcal{V}_2$. Then we have unique recovery in $\mathcal{V}$ under $\{i, \tau\}$ if and only if the following 4 conditions hold:*

$$\begin{aligned}
&(a1)\ \mathcal{V}_1 \cap \tau(\mathcal{V}_1) \subset \mathscr{E}_{\tau,1} \\
&(b1)\ \mathcal{V}_2 \cap \tau(\mathcal{V}_2) \subset \mathscr{E}_{\tau,1} \\
&(c1)\ \mathcal{V}_1 \cap \tau(\mathcal{V}_2 \setminus \mathcal{V}_1) = \varnothing \\
&(d1)\ \mathcal{V}_2 \cap \tau(\mathcal{V}_1 \setminus \mathcal{V}_2) = \varnothing
\end{aligned} \tag{4}$$

Proof. Having unique recovery in $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$ under $\{i, \tau\}$ with τ invertible is equivalent to:

$$\begin{aligned}
&(a2)\ \forall x \in \mathcal{V}_1, \forall y \in \mathcal{V}_1, x = \tau(y) \Rightarrow x = y \\
&(b2)\ \forall x \in \mathcal{V}_2, \forall y \in \mathcal{V}_2, x = \tau(y) \Rightarrow x = y \\
&(c2)\ \forall x \in \mathcal{V}_1, \forall y \in \mathcal{V}_2 \setminus \mathcal{V}_1, x = \tau(y) \Rightarrow x = y \\
&(d2)\ \forall x \in \mathcal{V}_2, \forall y \in \mathcal{V}_1 \setminus \mathcal{V}_2, x = \tau(y) \Rightarrow x = y
\end{aligned} \tag{5}$$

Now $(a2) \Leftrightarrow (a1)$, $(b2) \Leftrightarrow (b1)$ by Lemma 2 in [17]. Now $(c2)$ is equivalent to

$$\forall x \in \mathcal{V}_1, \forall y \in \mathcal{V}_2 \setminus \mathcal{V}_1, x \neq y \Rightarrow x \neq \tau(y)$$

For $x \in \mathcal{V}_1$ and $y \in \mathcal{V}_2 \setminus \mathcal{V}_1$, x cannot be equal to y so this is equivalent to:

$$\forall x \in \mathcal{V}_1, \forall y \in \mathcal{V}_2 \setminus \mathcal{V}_1, x \neq \tau(y)$$

which is equivalent to $(c1)$ $\mathcal{V}_1 \cap \tau(\mathcal{V}_2 \setminus \mathcal{V}_1) = \varnothing$. Similarly $(d2)$ is equivalent to $(d1)$ and we are done with the proof. □

When the input space becomes a union of two subspaces, we must have unique recovery under each subspace, and we also need to deal with the case where two different points that come from different subspaces may map to the same point after transformation, which is what $(c1)$ and $(d1)$ are about.

The empty set in the RHS of $(c1), (d1)$ may seem strange. An intuitive guess of simplification would be that it is equivalent to $\mathcal{V}_1 \cap \tau(\mathcal{V}_2) = 0$. However this is true only under certain conditions. A very simple counter-example would be when τ is the identity mapping, then $\mathcal{V}_1 \cap \tau(\mathcal{V}_2 \setminus \mathcal{V}_1) = \varnothing$ is always true, but $\mathcal{V}_1 \cap \tau(\mathcal{V}_2) = 0$ is not always true. In fact we have the following two conclusions:

- Suppose $\mathcal{V}_1 \cap \mathcal{V}_2 = 0$, then $\mathcal{V}_1 \cap \tau(\mathcal{V}_2 \setminus \mathcal{V}_1) = \varnothing \Rightarrow \mathcal{V}_1 \cap \tau(\mathcal{V}_2) = 0$ but not the other way around: pick $y \neq 0, y \in \ker(\tau) \cap \mathcal{V}_2$ shows that $\mathcal{V}_1 \cap \tau(\mathcal{V}_2 \setminus \mathcal{V}_1)$ contains at least 0.
- Suppose $\mathcal{V}_1 \cap \mathcal{V}_2 = 0$ and $\mathcal{V}_2 \cap \ker(\tau) = 0$, then $\mathcal{V}_1 \cap \tau(\mathcal{V}_2 \setminus \mathcal{V}_1) = \varnothing \Leftrightarrow \mathcal{V}_1 \cap \tau(\mathcal{V}_2) = 0$.

The proofs of these are rather trivial and are actually done implicitly in Lemma 3 and 4 which will come later, so they are omitted here.

When the subspaces are taken generically and τ is diagonalizable, it turns out that no extra conditions need to be added to deal with the case where different points from different subspaces may map to the same point, in other words, if the conditions of unique recovery in [17] for each of the two subspaces satisfy, we can have unique recovery under the union of these two subspaces.

Lemma 3 (Extension of Lemma 3 in [17])**.** *Let τ be a diagonalizable automorphism of $\mathbb{C}^m$ and $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$ union of two generic subspaces, $\dim(\mathcal{V}) \leq \lfloor m/2 \rfloor$. We have unique recovery in $\mathcal{V}$ under $\{i, \tau\}$ if and only if $\dim(\mathcal{V}) \leq \operatorname{codim}(\cup_{\lambda \neq 1} \mathscr{E}_{\tau,\lambda})$.*

Proof. $(\Rightarrow)$ By Lemma 2 we have $\mathcal{V}_1 \cap \tau(\mathcal{V}_1) \subset \mathscr{E}_{\tau,1}$ and $\mathcal{V}_2 \cap \tau(\mathcal{V}_2) \subset \mathscr{E}_{\tau,1}$, which implies $\dim(\mathcal{V}) \leq \operatorname{codim}(\cup_{\lambda \neq 1} \mathscr{E}_{\tau,\lambda})$. $(\Leftarrow)$ Same as in Lemma 2, we need to prove $(a2), (b2), (c2), (d2)$ to show unique recovery. Since $\dim(\mathcal{V}_1) \leq \dim(\mathcal{V})$, $\dim(\mathcal{V}_2) \leq \dim(\mathcal{V})$, by Lemma 3 in [17] $(a2), (b2)$ already hold. Using the same argument in Lemma 2, showing $(c2), (d2)$ is equivalent as showing $(c1), (d1)$ in (4). By Lemma 1, $\dim(\mathcal{V}_1 \cap \mathcal{V}_2) = 0$, thus $\mathcal{V}_1 \cap \mathcal{V}_2 = 0$ and therefore $\mathcal{V}_2 \setminus \mathcal{V}_1 = \mathcal{V}_2 \setminus \{0\}$. Now to show $(c1)$ $\mathcal{V}_1 \cap \tau(\mathcal{V}_2 \setminus \mathcal{V}_1) = \varnothing$, we need to show that $\mathcal{V}_1 \cap \tau(\mathcal{V}_2 \setminus \{0\}) = \varnothing$. Since τ is invertible, $\tau(\mathcal{V}_2 \setminus \{0\}) = \tau(\mathcal{V}_2) \setminus \{0\}$ and we need to show $\mathcal{V}_1 \cap (\tau(\mathcal{V}_2) \setminus \{0\}) = \varnothing$, which is apparently equivalent to showing $\mathcal{V}_1 \cap \tau(\mathcal{V}_2) = 0$. Since $\dim(\tau(\mathcal{V}_2)) \leq \dim(\mathcal{V}_2)$ and $\mathcal{V}_1$ is generic, by Lemma 1 we have $\dim(\mathcal{V}_1 \cap \tau(\mathcal{V}_2)) \leq \dim(\mathcal{V}_2) + \dim(\mathcal{V}_1) - m \leq 0$. Therefore $\mathcal{V}_1 \cap \tau(\mathcal{V}_2) = 0$. □

To ensure unique recovery when there're multiple automorphisms and union of multiple subspaces, we can require the corresponding conditions for each pair of automorphisms and each pair of subspaces to hold, which follows from Lemma 3 and the fact that the intersection of finitely many non-empty open sets is non-empty and open.

Proposition 1. Let $\mathcal{T}$ be a finite set of automorphisms of $\mathbb{C}^m$ with $\tau_1^{-1}\tau_2$ diagonalizable $\forall \tau_1, \tau_2 \in \mathcal{T}$. Let $\mathcal{V} = \mathcal{V}_1 \cup ... \cup \mathcal{V}_l$ be a union of multiple generic subspaces with $\dim(\mathcal{V}) \leq \lfloor m/2 \rfloor$. We have unique recovery in $\mathcal{V}$ under $\mathcal{T}$ if and only if $\dim(\mathcal{V}) \leq \operatorname{codim}(\cup_{\lambda \neq 1} \mathscr{E}_{\tau,\lambda})$ for every $\tau = \tau_1^{-1}\tau_2$ with $\tau_1, \tau_2 \in \mathcal{T}$.

Proof. $(\Rightarrow)$ Pick any $\tau_1, \tau_2 \in \mathcal{T}$ and any $v_1, v_2 \in \mathcal{V}_i$ for all i gives $\dim(\mathcal{V}_i) \leq \operatorname{codim}(\cup_{\lambda \neq 1} \mathscr{E}_{\tau,\lambda})$ for all i, so $\dim(\mathcal{V}) \leq \operatorname{codim}(\cup_{\lambda \neq 1} \mathscr{E}_{\tau,\lambda})$ for any τ_1, τ_2. $(\Leftarrow)$ Suppose $\tau_1(v_i) = \tau_2(v_j)$ for $v_i, v_j \in \mathcal{V}_i, \mathcal{V}_j$ correspondingly. If $i = j$ then this reduces to unique recovery in a single subspace, which by Lemma 3 in [17], holds true. Now suppose $i \neq j$. Since τ_1 is invertible, $v_i = \tau_1^{-1}\tau_2(v_j)$ and by Lemma 3 unique recovery holds. □

2.2 Sensing under Endomorphisms

When τ is not invertible, the argument in Lemma 3 is not correct any more, i.e., $\tau(\mathcal{V}_2 \setminus \{0\}) \neq \tau(\mathcal{V}_2) \setminus \{0\}$ since now $\ker(\tau)$ contains not only 0. This is handled by the fact that each subspace is chosen generically, in which case the existing conditions ensure that there is an open set from which each subspace intersects with $\ker(\tau)$ at only 0. Details are shown in the proof.

Lemma 4 (Extension of Lemma 4 in [17])**.** *Let $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$ be a union of two generic subspaces, with $\dim(\mathcal{V}) \leq \lfloor m/2 \rfloor$, and τ any endomorphism of $\mathbb{C}^m$ for which $\dim(\mathcal{V}) \leq \operatorname{codim}(\cup_{\lambda \neq 1} \mathscr{E}_{\tau,\lambda})$. Then we have unique recovery in $\mathcal{V}$ under $\{i, \tau\}$.*

Proof. Again, we need to show $(a2), (b2), (c2), (d2)$ in (5) to show unique recovery. According to Lemma 4 in [17], $(a2), (b2)$ already hold. $(c2), (d2)$ are equivalent to $(c1), (d1)$. Same as in Lemma 3, $\mathcal{V}_1 \cap \mathcal{V}_2 = 0$ and $\mathcal{V}_1 \cap \tau(\mathcal{V}_2) = 0$. To show $(c1)$ we need to show $\mathcal{V}_1 \cap \tau(\mathcal{V}_2 \setminus \{0\}) = \varnothing$. Suppose this is not true, then there exist $x \in \mathcal{V}_1, y \in \mathcal{V}_2, y \neq 0$ s.t. $\tau(y) = x$. Thus $x \in \tau(\mathcal{V}_2) \cap \mathcal{V}_1$, which gives $x = 0$. Now $\tau(y) = 0$ so $y \in \ker(\tau) \cap \mathcal{V}_2$. Since $\dim(\mathcal{V}_2) \leq \operatorname{codim}(\mathscr{E}_{\tau,0}) = m - \dim(\ker(\tau))$, by Lemma 1, $\dim(\ker(\tau) \cap \mathcal{V}_2) = 0$ so $\ker(\tau) \cap \mathcal{V}_2 = 0$ and therefore $y = 0$, which reaches a contradiction. Symmetrically we can show $(d1)$. □

Using Lemma 4 we can easily extend Theorem 1 in [17]. Same as in [17], define

$$\mathcal{Y}_{\rho\tau_1,\tau_2} = \{w : \dim(\operatorname{Span}(\rho\tau_1(w), \tau_2(w))) \leq 1\} \tag{6}$$

$$\mathcal{Z}_{\rho\tau_1,\tau_2} = \ker(\tau_2) \cup \ker(\rho\tau_1) \cup \ker(\rho\tau_1 - \tau_2) \tag{7}$$

$$\mathcal{U}_{\rho\tau_1,\tau_2} = \mathcal{Y}_{\rho\tau_1,\tau_2} \setminus \mathcal{Z}_{\rho\tau_1,\tau_2} \tag{8}$$

We have the following conclusion.

Theorem 1 (Extension of Theorem 1 in [17])**.** *Let $\mathcal{V} = \mathcal{V}_1 \cup ... \cup \mathcal{V}_l$ be a union of finitely many generic subspaces, with $\dim(\mathcal{V}) = n$. Let $\mathcal{T}$ be a finite set of endomorphisms of $\mathbb{C}^m$ where $m \geq 2n$ and for every $\tau \in \mathcal{T}$ we have $rank(\tau) \geq 2n$. Then we have unique recovery in $\mathcal{V}$ under $\mathcal{T}$ as long as for every $\tau_1, \tau_2 \in \mathcal{T}$ there is a projection ρ onto $\operatorname{im}(\tau_2)$ such that $\dim(\mathcal{V}) \leq \operatorname{codim}(\mathcal{U}_{\rho\tau_1,\tau_2})$.*

Proof. The proof is in spirit similar to the proof of Theorem 1 in [17]. Firstly it's enough to consider the case $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$ and $\mathcal{T} = \{\tau_1, \tau_2\}$. Let $\dim(\mathcal{V}_1) = n_1$ and $\dim(\mathcal{V}_2) = n_2$, then $n_1 \leq n, n_2 \leq n$. Suppose for $x, y \in \mathcal{V}, \tau_1(x) = \tau_2(y)$. Let $k = \operatorname{rank}(\tau_2)$. Since $k \geq 2n_1$ and $k \geq 2n_2$, we can view $\mathcal{V}_1$ and $\mathcal{V}_2$ as two generic subspaces of a generic k-dimensional subspace $\mathcal{H}$. Now $\rho\tau_1|_{\mathcal{H}}(x) = \tau_2|_{\mathcal{H}}(y)$ where ρ is the projection onto $\operatorname{im}(\tau_2)$. Since $\mathcal{H}$ is general we have $\mathcal{H} \cap \ker(\tau_2) = 0$ and so $\tau_2|_{\mathcal{H}}$ builds an isomorphism between $\mathcal{H}$ and $\operatorname{im}(\tau_2)$, and therefore $(\tau_2|_{\mathcal{H}})^{-1}\rho\tau_1|_{\mathcal{H}}(x) = y$. Let $\tau_{\mathcal{H}} = (\tau_2|_{\mathcal{H}})^{-1}\rho\tau_1|_{\mathcal{H}}$. By Lemma 4 we have unique recovery if $\dim(\mathscr{E}_{\tau_H,\lambda\neq 1}) \leq k - n$. Since $\mathscr{E}_{\tau_{\mathcal{H}},\lambda}$ is isomorphic to $\mathscr{E}_{(\rho\tau_1,\tau_2),\lambda\neq 1} \cap \mathcal{H}$ via $\tau_2|_{\mathcal{H}}$, this is equivalent as showing $\dim(\mathscr{E}_{(\rho\tau_1,\tau_2),\lambda\neq 1} \cap \mathcal{H}) \leq k - n$. For $\lambda \neq 0, 1$, $(\mathscr{E}_{(\rho\tau_1,\tau_2),\lambda} \cap \mathcal{H}) \subset (\mathcal{U}_{\rho\tau_1,\tau_2} \cap \mathcal{H}) \subset (\bar{\mathcal{U}}_{\rho\tau_1,\tau_2} \cap \mathcal{H})$, by Lemma 1 $\dim(\mathscr{E}_{(\rho\tau_1,\tau_2),\lambda} \cap \mathcal{H}) \leq k - n$. Now what's left to show is the case $\lambda = 0$. If $\dim(\mathscr{E}_{(\rho\tau_1,\tau_2),0}) \leq m - n$ we are done because $\mathcal{H}$ is general of dimension k, thus $\mathcal{H}$ intersects $\mathscr{E}_{(\rho\tau_1,\tau_2),0}$ at most at dimension $[(m-n)+k] - m = k - n$. If $\dim \mathscr{E}_{(\rho\tau_1,\tau_2),0} = \dim(\ker(\rho\tau_1)) \geq m - n$, then $\operatorname{im}(\tau_2)$ will intersect any complement C of $\operatorname{im}(\rho\tau_1)$ at dimension at least $[(m-n)+k] - m = k - n \geq n$. Let σ be the projection onto $C \cap \operatorname{im}(\tau_2)$. Applying σ on both sides of $\rho\tau_1(x) = \tau_2(y)$ gives $0 = \sigma\tau_2(y)$, so $y \in \ker(\sigma\tau_2) \cap \mathcal{V}$. Since $\dim(\ker(\sigma\tau_2)) \leq m - n$, we have $\mathcal{V}_1 \cap \ker(\sigma\tau_2) = 0$ and $\mathcal{V}_2 \cap \ker(\sigma\tau_2) = 0$, so that $y = 0$. Then $\tau_1(x) = \tau_2(y)$ gives $\tau_1(x) = 0$. Then $x \in \ker(\tau_1) \cap \mathcal{V}$. Since $\dim(\ker(\tau_1)) \leq m - n$, $\ker(\tau_1) \cap \mathcal{V}_1 = 0$ and $\ker(\tau_1) \cap \mathcal{V}_2 = 0$ so $x = 0$. □

Similarly when we consider unique recovery of generic vectors in $\mathcal{V}$, we have the following result in the union case.

Theorem 2 (Extension of Theorem 2 in [17])**.** *Let $\mathcal{T}$ be a finite set of endomorphisms of $\mathbb{C}^m$. Then a generic point v inside a union of finitely many generic subspaces $\mathcal{V} = \mathcal{V}_1 \cup ... \cup \mathcal{V}_l$, $\dim(\mathcal{V}) = n$, is uniquely recoverable in $\mathcal{V}$ under $\mathcal{T}$ as long as 1) $\operatorname{rank}(\tau) \geq n + 1$ for every $\tau \in \mathcal{T}$, and 2) no two endomorphisms in $\mathcal{T}$ are a scalar multiple of each other.*

Proof. It's enough to consider the case $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$ and $\mathcal{T} = \{\tau_1, \tau_2\}$. Let $\dim(\mathcal{V}_1) = n_1, \dim(\mathcal{V}_2) = n_2$, then $n_1 \leq n, n_2 \leq n$. We need to show the following where "GP" stands for "generic point".

$$(a) \text{ for GP } x \in \mathcal{V}_1, \forall y \in \mathcal{V}_1, \tau_1(x) = \tau_2(y) \Rightarrow x = y$$
$$(b) \text{ for GP } x \in \mathcal{V}_2, \forall y \in \mathcal{V}_2, \tau_1(x) = \tau_2(y) \Rightarrow x = y$$
$$(c) \text{ for GP } x \in \mathcal{V}_1, \forall y \in \mathcal{V}_2 \setminus \mathcal{V}_1, \tau_1(x) = \tau_2(y) \Rightarrow x = y$$
$$(d) \text{ for GP } x \in \mathcal{V}_2, \forall y \in \mathcal{V}_1 \setminus \mathcal{V}_2, \tau_1(x) = \tau_2(y) \Rightarrow x = y$$

By Theorem 2 in [17], $(a), (b)$ already hold. (c) is equivalent to

$$\text{for generic point } x \in \mathcal{V}_1, \forall y \in \mathcal{V}_2 \setminus \mathcal{V}_1, \tau_1(x) \neq \tau_2(y)$$

Suppose $\tau_1(x) = \tau_2(y)$. Let $V_1 \in \mathbb{C}^{m \times n_1}$ be a basis for $\mathcal{V}_1$ and $V_2 \in \mathbb{C}^{m \times n_2}$ be a basis for $\mathcal{V}_2$. Since $\tau_1(x) = \tau_2(y)$ we have $\tau_1(x) \in \tau_2(\mathcal{V}_2 \setminus \mathcal{V}_1) \subset \tau_2(\mathcal{V}_2)$. Let $x = V_1\xi$, where ξ is generic and $\xi \in \mathbb{C}^{n_1}$. Then $\operatorname{rank}([T_2V_2 \;\; T_1V_1\xi]) \leq n_2$, which cannot happen for generic $\mathcal{V}_1, \mathcal{V}_2, \xi$. This is because the $\mathcal{V}_1, \mathcal{V}_2, \xi$ that makes $\operatorname{rank}([T_2V_2 \;\; T_1V_1\xi]) = n_2 + 1$ forms a non-empty open set of $\operatorname{Gr}(n_1, m) \times \operatorname{Gr}(n_2, m) \times \mathbb{C}^{n_1}$. First, it forms an open set defined by the complement of the vanishing locus of all $(n_2 + 1) \times (n_2 + 1)$ minors of $[T_2V_2 \;\; T_1V_1\xi]$. Second, it's non-empty shown in the following. Since V_1, ξ are generic, $T_1V_1\xi$ is a generic vector in $R(T_1)$, which has dimension no less than $n + 1$. Since $\dim(\operatorname{Span}(T_2V_2)) \leq n_2 \leq n < \dim(R(T_1))$, $R(T_1) \setminus \operatorname{Span}(T_2V_2)$ is non-empty for any choice of V_2. Pick some $V_1\xi$ such that $T_1V_1\xi \in R(T_1) \setminus \operatorname{Span}(T_2V_2)$, then we have $\operatorname{rank}([T_2V_2 \;\; T_1V_1\xi]) = n_2 + 1$, so this set is non-empty. Similarly we can show (d). □

2.3 Uniqueness Guarantees for Sparse Signal Recovery

Here we develop theoretical guarantees for a class of *sparse signal recovery* problems. We say a point $x \in \mathbb{C}^n$ is a k-sparse vector if $\|x\|_0 \leq k$, where $\|\cdot\|$ counts the number of non-zero entries of a vector. We also say x is a generic k-sparse vector if it's k-sparse and the k nonzero entries of x are generic[4]. Under the previous notations, we are interested in the following problem: under what conditions do we have that, for k-sparse/generic k-sparse vectors x_1, x_2 and a generic matrix $A \in \mathbb{C}^{m \times n}$, whenever $T_1Ax_1 = T_2Ax_2$ then we have $x_1 = x_2$?

One important observation that we also discussed in the introduction is that for a k-sparse vector x, Ax is in the subspace spanned by the k columns of A indexed by the k nonzero entries of x. Therefore when we don't know which k entries of x are nonzero, Ax is in the *union* of $\binom{n}{k}$ k-dimensional subspaces, each spanned by k colomns of A. However, we can't apply the conclusion developed in previous sections directly, since these $\binom{n}{k}$ subspaces are not "mutually generic", in the sense that they may intersect non-trivially by including common column vectors of A. Specifically, suppose that n is big enough and $k = 2$. Then the subspace spanned by $A_{:,1}$, $A_{:,2}$ and the subspace spanned by $A_{:,1}$, $A_{:,3}$ both contain $A_{:,1}$, a case not covered by previous results. However this is a minor problem which can be fixed by the following two lemmas.

Lemma 5. *Let $\mathcal{W}_1, \mathcal{W}_2, \mathcal{W}_3$ be three generic subspaces in $\mathbb{C}^m$ with dimension d_1, d_2, d_3 where $d_1 \geq 1, d_2 \geq 1$, $d_1 + d_3 \leq m, d_2 + d_3 \leq m$. Let $\mathcal{V}_1 = \mathcal{W}_1 + \mathcal{W}_3$, $\mathcal{V}_2 = \mathcal{W}_2 + \mathcal{W}_3$ and $\mathcal{V} = \mathcal{V}_1 \cup \mathcal{V}_2$. Let $\mathcal{T}$ be a finite set of endomorphisms of $\mathbb{C}^m$ where $\operatorname{rank}(\tau) \geq 2(\max(d_1, d_2) + d_3)$ for every $\tau \in \mathcal{T}$. Then we have unique recovery of $\mathcal{V}$ under $\mathcal{T}$ as long as for every $\tau_1, \tau_2 \in \mathcal{T}$ there is a projection ρ onto $\operatorname{im}(\tau_2)$ such that $\dim(\mathcal{V}) \leq \operatorname{codim}(\mathcal{U}_{\rho\tau_1,\tau_2})$.*

Proof. We can first prove a "fixed" version of Lemma 4 where $\mathcal{V}$ has the new formulation here. We need to show $(a2)$, $(b2)$, $(c2)$ and $(d2)$. $(a2)$ and $(b2)$ hold by Lemma 4 in [17]. $(c2)$ and $(d2)$ are equivalent to $(c1)$ and $(d1)$, and also since they are symmetric, we only show $(c1)$ in the following. First since $\mathcal{V}_1 \cap \mathcal{V}_2 = \mathcal{W}_3$, we now need to show $\mathcal{V}_1 \cap \tau(\mathcal{V}_2 \setminus \mathcal{W}_3) = \varnothing$. Suppose this is not true, then there exist $v_1 \in \mathcal{V}_1, v_2 \in \mathcal{V}_2 \setminus \mathcal{W}_3$ such that $v_1 = \tau(v_2)$. Now let $v_1 = w_1 + w_3', v_2 = w_2 + w_3$ where $w_1 \in \mathcal{W}_1$, $w_2 \in \mathcal{W}_2$, $w_3, w_3' \in \mathcal{W}_3$ and $w_2 \neq 0$. Then $w_1 + w_3' = \tau(w_2) + \tau(w_3)$. This implies $w_1 \in \mathcal{W}_1 \cap (\tau(\mathcal{W}_2) + \tau(\mathcal{W}_3) + \mathcal{W}_3)$, which has dimension at most $d_1 + d_2 + 2d_3 - m = \dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) - m \leq 0$, so $w_1 = 0$. Therefore $\tau(w_2 + w_3) = w_3'$. Now note that both $w_2 + w_3$ and w_3' are vectors in $\mathcal{V}_2$ for which we have shown that all vectors are uniquely recoverable,

[4] we say k nonzero entries here instead of "less or equal than k nonzero entries" since the instances of x that has less than k nonzero entries are in a set with measure zero.

so $w_2 + w_3 = w_3'$. This implies $w_2 \in \mathcal{W}_2 \cap \mathcal{W}_3$, so $w_2 = 0$ which reaches a contradiction. Here the proof of the "fixed" version of Lemma 4 is finished. Now follow the proof of Theorem 1 will finish the proof of this proposition, where two changes in the proof are made: first, change $\mathcal{H}$ to be a generic subspace of dimension $d_1 + d_2 + d_3$ which contains $\mathcal{V}_1$ and $\mathcal{V}_2$; second, use the "fixed" version of Lemma 4 which we described above instead of Lemma 4. □

Lemma 6. *Let $\mathcal{W}_1, \mathcal{W}_2, \mathcal{W}_3$ be three generic subspaces in $\mathbb{C}^m$ with dimension d_1, d_2, d_3 where $d_1 \geq 1, d_2 \geq 1$, $d_1 + d_3 \leq m, d_2 + d_3 \leq m$. Let $\mathcal{V}_1 = \mathcal{W}_1 + \mathcal{W}_3$, $\mathcal{V}_2 = \mathcal{W}_2 + \mathcal{W}_3$. Let $\mathcal{T}$ be a finite set of endomorphisms of $\mathbb{C}^m$. Then any generic point inside $\mathcal{V}_1 \cup \mathcal{V}_2$ is uniquely recoverable under $\mathcal{T}$ as long as 1) $\operatorname{rank}(\tau) \geq \max(d_1, d_2) + d_3 + 1$ for every $\tau \in \mathcal{T}$, and 2) no two endomorphisms in $\mathcal{T}$ are a scalar multiple of each other.*

Proof. We need to show the following where "GP" stands for "generic point". Here τ_1 and τ_2 may equal.

$$(a) \text{ for GP } x \in \mathcal{V}_1, \forall y \in \mathcal{V}_1, \tau_1(x) = \tau_2(y) \Rightarrow x = y$$
$$(b) \text{ for GP } x \in \mathcal{V}_2, \forall y \in \mathcal{V}_2, \tau_1(x) = \tau_2(y) \Rightarrow x = y$$
$$(c) \text{ for GP } x \in \mathcal{V}_1, \forall y \in \mathcal{V}_2 \setminus \mathcal{V}_1, \tau_1(x) = \tau_2(y) \Rightarrow x = y$$
$$(d) \text{ for GP } x \in \mathcal{V}_2, \forall y \in \mathcal{V}_1 \setminus \mathcal{V}_2, \tau_1(x) = \tau_2(y) \Rightarrow x = y$$

By Theorem 2 in "HS", $(a), (b)$ already hold. (c) is equivalent to showing

$$(e) \text{ for generic point } x \in \mathcal{V}_1, \forall y \in \mathcal{V}_2 \setminus \mathcal{V}_1, \tau_1(x) \neq \tau_2(y)$$

First suppose $\tau_1 \neq \tau_2$. We'll show that in this case $\tau_1(x) = \tau_2(y)$ cannot happen. Suppose $\tau_1(x) = \tau_2(y)$. Let W_i be a basis for $\mathcal{W}_i$, then $\mathcal{V}_1$ would have basis $V_1 = [W_1; W_3]$ and $\mathcal{V}_2$ has basis $V_2 = [W_2; W_3]$. Since $\tau_1(x) = \tau_2(y)$ we have $\tau_1(x) \in \tau_2(\mathcal{V}_2 \setminus \mathcal{V}_1) \subset \tau_2(\mathcal{V}_2)$. Let $x = V_1\xi$, where ξ is generic and $\xi \in \mathbb{C}^{d_1+d_3}$. Then $[T_2V_2 \;\; T_1V_1\xi]$ doesn't have full column rank, which cannot happen for generic $\mathcal{W}_1, \mathcal{W}_2, \mathcal{W}_3, \xi$. This is because the $\mathcal{W}_1, \mathcal{W}_2, \mathcal{W}_3, \xi$ that makes $[T_2V_2 \;\; T_1V_1\xi]$ full column rank forms a non-empty open set of $\operatorname{Gr}(d_1, m) \times \operatorname{Gr}(d_2, m) \times \operatorname{Gr}(d_3, m) \times \mathbb{C}^{d_1+d_3}$. First, it forms an open set defined by the complement of the vanishing locus of all $(d_2 + d_3 + 1) \times (d_2 + d_3 + 1)$ minors of $[T_2V_2 \;\; T_1V_1\xi]$. Second, it's non-empty shown in the following. Let $v \in \mathbb{C}^m$ s.t. $\tau_1(v)$ and $\tau_2(v)$ are linearly independent. Let C be a complement of $\operatorname{im}(\tau_2) \cap \operatorname{Span}(\tau_1(v), \tau_2(v))$ in $\operatorname{im}(\tau_2)$. Then $\dim(C) \geq \max(d_1, d_2) + d_3 - 1$. Since $d_1 \geq 1, d_2 \geq 1$, we have $\dim(C) \geq d_1 - 1, \dim(C) \geq d_2 - 1, \dim(C) \geq d_3$. Therefore we can pick $\mathcal{U}_1, \mathcal{U}_2, \mathcal{U}_3 \subset C$ s.t. $\dim(\mathcal{U}_1) = d_1 - 1$, $\dim(\mathcal{U}_2) = d_2 - 1$, $\dim(\mathcal{U}_3) = d_3$. Let $\mathcal{W}_1', \mathcal{W}_2', \mathcal{W}_3$ be $d_1 - 1, d_2 - 1, d_3$ dimensional subspaces s.t. $\tau_2(\mathcal{W}_1') = \mathcal{U}_1, \tau_2(\mathcal{W}_2') = \mathcal{U}_2, \tau_2(\mathcal{W}_3) = \mathcal{U}_3$. Now let $\mathcal{W}_1 = \mathcal{W}_1' + \operatorname{Span}(v)$ and $\mathcal{W}_2 = \mathcal{W}_2' + \operatorname{Span}(v)$. Now we have that $v \in \mathcal{W}_1 + \mathcal{W}_3$ and $\dim(\tau_1(v) + \tau_2(\mathcal{W}_2 + \mathcal{W}_3)) = d_2 + d_3 + 1$. Similarly we can show (d).

Now suppose $\tau_1 = \tau_2$. Then $x - y \in \ker(\tau_1)$. Let $x = w_1 + w_3^{(1)}$ and $y = w_2 + w_3^{(2)}$, where $w_1 \in \mathcal{W}_1, w_2 \in \mathcal{W}_2, w_3^{(1)}, w_3^{(2)} \in \mathcal{W}_3$ and $w_1, w_3^{(1)}$ are generic. Then $w_1 \in \ker(\tau_1) + \mathcal{W}_2 + \mathcal{W}_3$. This cannot happen since $\dim(\ker(\tau_1) + \mathcal{W}_2 + \mathcal{W}_3) < m$, and w_1 is generic in $\mathcal{W}_1$. Similarly we can show (d). □

Extension of the above results to the case of multiple union and finitely many transformations then follows naturally.

Specializing the above results to sparse signal recoveries, we have the following theorem.

Theorem 3. *Let $\mathcal{T}$ be a set of matrices in $\mathbb{C}^{m\times m}$. Let $A \in \mathbb{C}^{m\times n}$ be a general matrix. Suppose we have k-sparse vectors $x_1, x_2 \in \mathbb{C}^n$ and $T_1, T_2 \in \mathcal{T}$. Then:*

- *If $\operatorname{rank}(T) \geq 2k$ for every $T \in \mathcal{T}$ and for every $T_1, T_2 \in \mathcal{T}$ there is a projection P onto $\mathcal{R}(T_2)$ such that $dim(\mathcal{U}_{PT_1,T_2}) \leq m - k$, then $T_1Ax_1 = T_2Ax_2$ implies $x_1 = x_2$.*
- *When x_1, x_2 are generic k-sparse vectors, if each element in $\mathcal{T}$ has rank at least $k + 1$ and no two elements are multiple of each other, then $T_1Ax_1 = T_2Ax_2$ implies $x_1 = x_2$.*

Proof. If the index set for the non-zero entries of a k-sparse vector x is I, then Ax is a vector in $\mathcal{R}(A_{:,I})$, which is the Span of the columns of A indexed by I and its dimension is k. Suppose

$T_1Ax_1 = T_2Ax_2$. Let the index set of x_1 be I and the index set of x_2 be J. If $I = J$ then the problem reduces to the single subspace model, therefore the two claims holds by Theorem 1 and Theorem 2 in [17], so $Ax_1 = Ax_2$. Let the vector of nonzero entries indexed by I in x_1, x_2 be ξ_1, ξ_2 respectively, then $A_{:,I}\xi_1 = A_{:,I}\xi_2$, which implies $\xi_1 = \xi_2$ since $m \leq k$. Now $\xi_1 = \xi_2$ and $I = J$ gives $x_1 = x_2$. In what follows, we discuss the case where $I \neq J$. Suppose $|I \cap J| = d$ where $0 \leq d \leq k-1$. Let $\mathcal{W}_1 = \mathcal{R}(A_{:,I\setminus J})$, $\mathcal{W}_2 = \mathcal{R}(A_{:,J\setminus I})$ and $\mathcal{W}_3 = \mathcal{R}(A_{:,I\cap J})$, then $\dim(\mathcal{W}_1) = \dim(\mathcal{W}_2) = k - d \geq 1, \dim(\mathcal{W}_3) = d$. Now we separately prove the two claims.

- **First claim.** Now $Ax_1 \in \mathcal{W}_1 + \mathcal{W}_2$ and $Ax_2 \in \mathcal{W}_2 + \mathcal{W}_3$. By Lemma 5 we have that $T_1Ax_1 = T_2Ax_2$ implies $Ax_1 = Ax_2$. Now, $x_1 - x_2$ can have at most $2k$ non-zero entries, in other words, $x_1 - x_2$ is inside a subspace with dimension at most $2k$ (in fact, the subspace spanned by the standard canonical vectors indexed by $I \cup J$). Call this subspace U. Now $x_1 - x_2 \in U \cap \mathcal{N}(A)$. Since A is generic, the dimension of $U \cap \mathcal{N}(A)$ is at most $2k + n - m - n = 2k - m \leq 0$, and therefore this intersection is 0 which implies $x_1 = x_2$.
- **Second claim.** Ax_1 is a generic vector in $\mathcal{W}_1 + \mathcal{W}_2$ and $Ax_2 \in \mathcal{W}_2 + \mathcal{W}_3$. Since $\max(\dim(\mathcal{W}_1), \dim(\mathcal{W}_2)) + \dim(\mathcal{W}_3) = k - d + d = k \leq \mathrm{rank}(T_i)$, by Lemma 6 we have that $T_1Ax_1 = T_2Ax_2$ implies $Ax_1 = Ax_2$. Now $x_1 - x_2 \in \mathcal{N}(A)$. Let the k-dimensional subspaces $\mathcal{U}_1, \mathcal{U}_2$ be subspaces spanned by the standard basis vectors for the index set I and J respectively (note that $\mathcal{U}_1 \neq \mathcal{U}_2$). Now $x_1 \in (\mathcal{U}_2 + \mathcal{N}(A))$, which cannot happen for generic x_1 in $\mathcal{U}_1$, since the set of (x_1, A) s.t. $x_1 \notin (\mathcal{U}_2 + \mathcal{N}(A))$ is an open set and we can exhibit x_1, A in the set by the following. First note that $\dim(\mathcal{U}_2 + \mathcal{N}(A)) \leq k + n - m \leq n - 1$. Pick $x_1 \in \mathcal{U}_1 \setminus \mathcal{U}_2$ and pick A s.t. $\mathcal{U}_2 + \mathcal{N}(A)$ is a subset of any complement of $\mathrm{Span}(x_1)$ we have $x_1 \notin (\mathcal{U}_2 + \mathcal{N}(A))$. Therefore $x_1 = x_2$.

□

Applying the above conclusions by specializing $\mathcal{V}$ and $\mathcal{T}$, we can extend many previous results into sparse signal case. We illustrate this using some examples.

Corollary 1. *In linear regression without correspondence, we have that $\mathcal{T}$ is the set of permutation matrices of size $m \times m$ and $\mathcal{V}$ is the column space of a generic matrix, say $A \in \mathbb{C}^{m\times n}$. If we have $m \geq 2k$, then any k-sparse vector $x \in \mathbb{C}^n$ is uniquely recoverable, in other words, $T_1Ax_1 = T_2Ax_2$ implies $x_1 = x_2$.*

Proof. Combining Proposition 3 in [17] and Theorem 3 we can obtain the desired conclusion. □

We also provide an application in real phase retrieval, where recovery can only be guaranteed up to sign due to the nature of the problem which also causes a bit more steps to prove.

Corollary 2. *In real phase retrieval, the set $\mathcal{T}$ consists of all the sign matrices (diagonal matrix with 1/-1 on the diagonal) and they all have full rank. When the ambient dimension is m, we have "unique recovery up to sign" for generic k-sparse vectors as long as $m \geq k + 1$.*

Proof. The only way two matrices $T_1, T_2 \in \mathcal{T}$ can be multiple of each other is $T_1 = T_2$ or $T_1 = -T_2$; in which case $T_1Ax_1 = T_2Ax_2$ implies $x_1 = x_2$ and $x_1 = -x_2$ correspondingly. Now we can consider the case T_1 and T_2 are different and not multiple of each other. Now follow the same procedure in Theorem 3 we have that $T_1Ax_1 = T_2Ax_2$ implies $x_1 = x_2$. Therefore we have "unique recovery up to sign" when $m \geq k + 1$. □

3 Noisy Homomorphic Sensing

In this section we study HS of a linear subspace under noise.

3.1 Modelling Assumptions

We first formulate here our model which takes noise into account. Starting from here we will stop using the abstract linear algebra notations and switch to matrices and coordinate vectors. We

will abuse the notations so that $\mathcal{T}$ refers to the set of matrix representations of the corresponding endomorphisms.

Let the noiseless model be $y = T^* A x^*$ for some $T^* \in \mathcal{T}$ and $x^* \in \mathbb{C}^n$, where A is a basis matrix for $\mathcal{V}$. We now corrupt y by some noise vector ϵ, so what we observe becomes $\bar{y} = y + \epsilon$. We also make the following assumption.

Assumption 1. A is a generic matrix, and the conditions for unique recovery (in noiseless case) of $\mathcal{V}$ under $\mathcal{T}$ are satisfied.

Now we study how the estimator $\hat{x}$ as following behaves under various scale of the noise vector ϵ:

$$(\hat{T}, \hat{x}) \in \underset{T \in \mathcal{T}, x \in \mathbb{C}^n}{\arg\min} ||\bar{y} - TAx||_2 \tag{9}$$

Although this estimator is equivalent to the Maximum Likelihood Estimator under the assumption that the noise is i.i.d Gaussian, here we don't make any statistical assumptions about ϵ.

3.2 Deterministic Results

First we define the non-empty set

$$\mathcal{T}_1 := \{T \in \mathcal{T} : y \in \mathcal{R}(TA)\} \tag{10}$$

This set is non-empty since it at least contains T^*. Utilizing the uniqueness guarantee assumption, we have the following conclusion.

Lemma 7. *Under Assumption 1, if $\hat{T} \in \mathcal{T}_1$, then $\hat{T}A(\hat{x} - x^*) = P_{\hat{T}A}\epsilon$.*

Proof. Since $\hat{T} \in \mathcal{T}_1$, there exists some $\tilde{x}$ s.t. $\hat{T}A\tilde{x} = y = T^*Ax^*$. Now by Assumption 1, this implies $\tilde{x} = x^*$ and therefore $y = \hat{T}Ax^*$. Now

$$\hat{x} = \arg\min_x ||\hat{T}Ax^* - \hat{T}Ax + \epsilon||_2 \tag{11}$$

and from here it can be easily seen that $\hat{T}A(\hat{x} - x^*) = P_{\hat{T}A}\epsilon$. □

Therefore it can be seen that we are in the "good case" if $\hat{T} \in \mathcal{T}$, and the question naturally becomes under what conditions do we achieve $\hat{T} \in \mathcal{T}_1$. We provide a sufficient condition in the following lemma.

Lemma 8. *We have $\hat{T} \in \mathcal{T}_1$ as long as*

$$2||\epsilon||_2 < \max_{T \in \mathcal{T} \setminus \mathcal{T}_1} (||y||_2 - ||P_{TA}y||_2) \tag{12}$$

Proof. We begin by rewriting $\hat{T}$ to be

$$\begin{aligned}
\hat{T} &= \arg\min_{T \in \mathcal{T}} \min_{w \in \mathcal{R}(TA)} ||\bar{y} - w||_2 \\
&= \arg\min_{T \in \mathcal{T}} \min_{w \in \mathcal{R}(TA)} (||w||_2^2 - 2\bar{y}^T w) \\
&= \arg\min_{T \in \mathcal{T}} \min_{\lambda > 0} \min_{w \in \mathcal{R}(TA), ||w||_2 = \lambda} (\lambda^2 - 2\bar{y}^T w) \\
&= \arg\min_{T \in \mathcal{T}} \min_{\lambda > 0} (\lambda^2 - 2\lambda \max_{w \in \mathcal{R}(TA), ||w||_2 = 1} \bar{y}^T w) \\
&= \arg\max_{T \in \mathcal{T}} \max_{w \in \mathcal{R}(TA), ||w||_2 = 1} \bar{y}^T w
\end{aligned} \tag{13}$$

Now it's enough to show that for any $T_2 \in \mathcal{T} \setminus \mathcal{T}_1$, there exists some $T_1 \in \mathcal{T}_1$ such that

$$\max_{w \in \mathcal{R}(T_1 A), ||w||_2 = 1} \bar{y}^T w > \max_{w \in \mathcal{R}(T_2 A), ||w||_2 = 1} \bar{y}^T w \tag{14}$$

which surely holds if

$$\bar{y}^T y/||y||_2 > \max_{w \in \mathcal{R}(T_2 A), ||w||_2=1} \bar{y}^T w \tag{15}$$

Now let w_2 be a maximizer of the right hand side, then this is equivalent to

$$||y||_2 > y^T w_2 + \epsilon^T w_2 - e^T y/||y||_2 \tag{16}$$

Now since $y^T w_2 \leq \max_{w \in \mathcal{R}(T_2 A), ||w||_2=1} y^T w$, $\epsilon^T w_2 \leq ||\epsilon||_2$ and $\epsilon^T y \leq ||\epsilon||_2 ||y||_2$, it suffices to have

$$||y||_2 > 2||\epsilon||_2 + \max_{w \in \mathcal{R}(T_2 A), ||w||_2=1} y^T w \tag{17}$$

which is equivalent to

$$2||\epsilon||_2 < ||y||_2 - ||P_{TA} y||_2 \tag{18}$$

Now adding the outer maximization over $\mathcal{T} \setminus \mathcal{T}_1$ we obtain the desired result. □

3.3 Probabilistic Results

In this part we perform probabilistic analysis of the deterministic results above. First we make the following two assumptions which will be based upon for the rest of this section.

Assumption 2. The n-dimensional linear subspace $\mathcal{V}$ is a uniformly random subspace in the Grassmannian $\mathrm{Gr}(m, n)$ according to the Haar measure.

Recall that the columns of A span $\mathcal{V}$. To say that $\mathcal{V}$ is uniformly random according to the Haar measure is to say that A has i.i.d. standard Gaussian elements.

Assumption 3. The singular values of every $T \in \mathcal{T}$ are 1 or 0.

Assumption 2 is a rather common assumption of the subspace $\mathcal{V}$ which is also adopted by some analysis of noise case in homomorphic sensing's instances such as in [16, 12, 4]. Assumption 3 seems like a strong assumption but is in fact satisfied by almost all existing applications of HS. For example,

- In Unlabelled Sensing, $\mathcal{T}$ consists of selection matrices whose singular values are 1 or 0.
- In linear regression without correspondences, $\mathcal{T}$ consists of permutation matrices whose singular values are 1.
- In real phase retrieval, $\mathcal{T}$ consists of sign matrices whose singular values are 1.
- Any composition of permutation, coordinate projection and sign corruptions starting with permutation would have 1/0 singular values.

Assumption 3 is also vital for probabilistic analysis under Assumption 2, since the image of a uniformly random subspace under transformations with 1/0 singular values is still a uniformly random subspace, which is important for subsequent probabilistic analysis.

Next we define the signal-to-noise ratio (snr) and the estimation error (err) of our model, whose relation will be studied later:

$$\text{snr} = \frac{||x^*||_2}{||\epsilon||_2} \tag{19}$$

$$\text{err} = \frac{||\hat{x} - x^*||_2}{||\epsilon||_2} \tag{20}$$

We now begin our analysis. From our deterministic results, we can see that the crucial step is to bound the probability of the event $\hat{\tau} \in \mathcal{T}$, after which we can bound the error $||x^* - \hat{x}||_2$ rather easily. Now by Lemma 8, we need to bound the probability of $2||\epsilon||_2 < \max_{T \in \mathcal{T} \setminus \mathcal{T}_1} (||y||_2 - ||P_{TA} y||_2)$. Since the outer maximization can be dealt with using union bound, we need to bound the probability of

$$2||\epsilon||_2 < ||y||_2 - ||P_{TA} y||_2 \tag{21}$$

for some $T \in \mathcal{T} \setminus \mathcal{T}_1$.

By rotational invariance of A, we can reduce the problem to the case $x^* = ||x^*||_2 e_1$ where e_1 is the first standard canonical vector, so $Ax^* = ||x^*||_2 a$ where $a = A_{:,1}$. Now we need to find a lower bound for the probability

$$P(\frac{2}{\text{snr}} < ||T^* a||_2 - ||P_{TA} T^* a||_2) \tag{22}$$

We now split our analysis into several stages.

3.3.1 Reduction to 1-Dimensional Case

Let us analyze $||P_{TA} T^* a||_2^2$ first, for which we have the following conclusion which reduces A to a.

Lemma 9. *Under Assumption 2 and Assumption 3,*

$$||P_{TA} T^* a||_2^2 = (1 - X)||T^T T^* a||_2^2 + X||P_{Ta} T^* a||_2^2 \tag{23}$$

Here $X = ||P_{r-n}^{r-1} v||_2^2$ is the squared norm of a random projection which is a r.v. independent of a and $0 \leq X \leq 1$, where v represents a fixed standard basis vector in $r - 1$ dimensions.

Proof. Let the Thin-SVD of T be $U_T \Sigma_T V_T^T = U_T V_T^T$, where $U_T, V_T \in \mathbb{R}^{m \times r}$. We will use the following simple lemma.

Lemma 10. *Let U, X be matrices and v be a vector, each with appropriate size. If $U^T U = I$, then*

$$||P_{UX} v||_2^2 = ||P_X U^T v||_2^2 \tag{24}$$

Proof. $P_{UX} v = UX((UX)^T UX)^{-1}(UX)^T v = UX(X^T X)^{-1} X^T U^T v = U P_X U^T v$. Since $\forall w, ||Uw||_2^2 = w^T U^T U w = ||w||_2^2$, we have the desired result. □

Using this lemma we have that $||P_{TA} T^* a||_2^2 = ||P_{V_T^T A} U_T^T T^* a||_2^2$. Now note that $V_T^T A \in \mathbb{R}^{r \times n}$ is an i.i.d Gaussian matrix, but we can't directly replace $V_T^T A$ by another independent matrix due to the existence of a. Here we use the idea in the proof of Lemma 5 in [12] to eliminate this dependence. First since $||P_{V_T^T A} U_T^T T^* a||_2^2 = ||U_T^T T^* a||_2^2 - ||P_{(V_T^T A)^\perp} U_T^T T^* a||_2^2$, we shift to the term $||P_{(V_T^T A)^\perp} U_T^T T^* a||_2^2$. Let $S_1 = \mathcal{R}(V_T^T a)$ and $S_{-1} = \mathcal{R}(V_T^T A_{:,2:n})$, then $\mathcal{R}(V_T^T A)^\perp = S_1^\perp \cap S_{-1}^\perp$. Now we have

$$\begin{aligned} ||P_{(V_T^T A)^\perp} U_T^T T^* a||_2^2 &= ||P_{S_1^\perp \cap S_{-1}^\perp} P_{S_1^\perp} U_T^T T^* a||_2^2 \\ &= ||P_{S_1^\perp \cap S_{-1}^\perp} u||_2^2 \cdot ||P_{S_1^\perp} U_T^T T^* a||_2^2 \end{aligned} \tag{25}$$

where $u = \frac{P_{S_1^\perp} U_T^T T^* a}{||P_{S_1^\perp} U_T^T T^* a||_2}$.

Now S_{-1} is the range space of a $r \times (n-1)$ i.i.d Gaussian matrix, so $S_{-1}^\perp$ is a uniformly random $r - n + 1$ dimensional subspace in $\mathbb{R}^r$. Conditioning on a, $S_1^\perp$ is a fixed $r - 1$ dimensional subspace, therefore $S_1^\perp \cap S_{-1}^\perp$ is a uniformly random $r - n$ dimensional subspace within $S_1^\perp$. Therefore $||P_{S_1^\perp \cap S_{-1}^\perp} u||_2^2$ is equivalent to $X = ||P_{r-n}^{r-1} v||_2^2 = 1 - ||P_{n-1}^{r-1} v||_2^2$, the squared norm of a random projection, where v represents a fixed standard basis vector in $r - 1$ dimensions. □

So now we need to find a lower bound for

$$P\left(\frac{2}{\text{snr}} < ||T^* a||_2 - \sqrt{(1 - X)||T^T T^* a||_2^2 + X||P_{Ta} T^* a||_2^2}\right) \tag{26}$$

We can further relax this term by

$$
\begin{aligned}
&P\left(\frac{2}{\text{snr}} < ||T^*a||_2 - \sqrt{(1-X)||T^TT^*a||_2^2 + X||P_{Ta}T^*a||_2^2}\right) \\
&\geq P\left(\frac{2}{\text{snr}} < ||T^*a||_2 - \sqrt{||T^*a||_2^2 - X(||T^*a||_2^2 - ||P_{Ta}T^*a||_2^2)}\right) \\
&\geq P\left(X > \frac{4}{\text{snr}} \cdot \frac{||T^*a||_2}{||T^*a||_2^2 - ||P_{Ta}T^*a||_2^2}\right)
\end{aligned} \tag{27}
$$

and therefore we now need to find an appropriate probabilistic upper bound for

$$
\frac{||T^*a||_2}{||T^*a||_2^2 - ||P_{Ta}T^*a||_2^2} \tag{28}
$$

3.3.2 A Further Relaxation (full rank case)

We analyze the case where T's have full rank, and also discuss about the general case where T's may be rank-deficient in Section 3.3.6.

When T's have full rank then all the singular values are 1, and therefore in fact T's are orthogonal matrices. Define

$$
d(T, T^*) = 1 - \frac{| < T, T^* > |}{m} \tag{29}
$$

as a measure of the "distance" between T, T^*. Note that $0 \leq d(T, T^*) \leq 1$. Now we have the following result.

Lemma 11. *Under Assumption 2 and Assumption 3, when T's are full rank, we have for $\delta_1, \delta_2 > 0$,*

$$
\frac{||T^*a||_2}{||T^*a||_2^2 - ||P_{Ta}T^*a||_2^2} \leq \frac{(1+\delta_2)}{\delta_1\sqrt{m} \cdot d(T, T^*)} \tag{30}
$$

w.p. at least $1 - Cm\delta_1^{1/2} - 2\exp(-Cm\delta_2^2)$.

Proof. For the denominator, we have

$$
\begin{aligned}
||T^*a||_2^2 - ||P_{Ta}T^*a||_2^2 &= ||a||_2^2 - \frac{(a^TT^TT^*a)^2}{||a||_2^2} \\
&\geq ||a||_2^2 - |a^TT^TT^*a|
\end{aligned} \tag{31}
$$

Let Λ be the diagonal eigenvalue matrix of $\frac{T^TT^* + T^{*T}T}{2}$. By rotational invariance of Gaussian vectors, $||a||_2^2 - |a^TT^TT^*a|$ is equivalent to $||a||_2^2 - |a^T\Lambda a|$ in distribution. For some positive value t, we now bound $P(||a||_2^2 - |a^T\Lambda a| \leq t)$. Using union bound for several times and Lemma 8 in [12] we have,

$$
\begin{aligned}
&P(||a||_2^2 - |a^T\Lambda a| \leq t) \\
&\quad \leq P(||a||_2^2 - a^T\Lambda a \leq t) + P(||a||_2^2 + a^T\Lambda a \leq t) \\
&\quad = P(a^T(I-\Lambda)a \leq t) + P(a^T(I+\Lambda)a \leq t) \\
&\quad \leq \sum_{i\in[m],\Lambda_i\neq 1} P(a_i^2 \leq \frac{t}{m - Tr(\Lambda)}) + \sum_{i\in[m],\Lambda_i\neq -1} P(a_i^2 \leq \frac{t}{m + Tr(\Lambda)}) \\
&\quad \leq 2m \cdot P(a_1^2 \leq \frac{t}{m - |Tr(\Lambda)|}) \\
&\quad \leq 2m \cdot (\delta\exp(1-\delta))^{1/2} \\
&\quad \leq Cm\delta^{1/2}
\end{aligned} \tag{32}
$$

where $t = \delta(m - |Tr(\Lambda)|) = \delta m \cdot d(T, T^*)$ and $0 < \delta < 1$. So now

$$||T^*a||_2^2 - ||P_{Ta}T^*a||_2^2 \geq \delta_1 m \cdot d(T, T^*) \tag{33}$$

w.p. at least $1 - Cm\delta_1^{1/2}$.

Now for the nominator $||T^*a||_2 = ||a||_2$, we have $||a||_2 \leq \sqrt{m}(1 + \delta_2)$ w.p. at least $1 - 2\exp(-Cm\delta_2^2)$ by Theorem 6.3.2 in [20]. Summarizing the bounds for both the denominator and the nominator we finish the proof. □

3.3.3 Applying Bounds on Random Projections

We now bound $P(X > \frac{4}{\text{snr}} \cdot \frac{(1+\delta_2)}{\delta_1\sqrt{m}d(T,T^*)})$, continuing Equation (27). Applying Lemma 10 in [12] with $\beta = (1 - z)\frac{r-1}{n-1}$ where z is chosen s.t. $\beta > 1$ (which is equivalent to $z < \frac{r-n}{r-1}$), we have

$$\begin{aligned} P(X < z) &\leq (\frac{r-1}{n-1})^{\frac{n-1}{2}} \cdot (\frac{r-1}{r-n})^{\frac{r-n}{2}} \cdot z^{\frac{r-n}{2}} \\ &\leq 2^{r-1} z^{\frac{r-n}{2}} \end{aligned} \tag{34}$$

Applying this directly with $z = \frac{4}{\text{snr}} \cdot \frac{(1+\delta_2)}{\delta_1\sqrt{m}d(T,T^*)}$ and further combined with Lemma 9, Equation (27) and Lemma 11 yields

Lemma 12. *Under Assumption 2 and Assumption 3, when T's are full rank, we have for $\delta_1, \delta_2 > 0$, if*

$$\textit{snr} > \frac{4(m-1)(1+\delta_2)}{(m-n)\delta_1\sqrt{m} \cdot d(T, T^*)} \tag{35}$$

Then $2||\epsilon||_2 < ||y||_2 - ||P_{TA}y||_2$ w.p. at least

$$1 - Cm\delta_1^{1/2} - 2\exp(-Cm\delta_2^2) - 2^{2m-n-1}(\frac{1+\delta_2}{\delta_1\sqrt{m}d(T,T^*)})^{\frac{m-n}{2}}(\frac{1}{\textit{snr}})^{\frac{m-n}{2}} \tag{36}$$

Now applying union bound on the set $\mathcal{T} \setminus T^*$ where we pick the same δ_1, δ_2 for each T and utilizing Lemma 8, we have

Proposition 2. Under Assumption 2 and Assumption 3, when T's are full rank, we have for $\delta_1, \delta_2 > 0$, if

$$\text{snr} > \frac{4(m-1)(1+\delta_2)}{(m-n)\delta_1\sqrt{m} \cdot \min_{T\in\mathcal{T}\setminus T^*} d(T, T^*)} \tag{37}$$

Then $\hat{\tau} \in \mathcal{T}_1$ w.p. at least

$$\begin{aligned} &1 - |\mathcal{T}|(Cm\delta_1^{1/2} + 2\exp(-Cm\delta_2^2)) \\ &- 2^{2m-n-1}\left(\frac{1+\delta_2}{\delta_1\sqrt{m}}\right)^{\frac{m-n}{2}}\left(\sum_{T\in\mathcal{T}\setminus T^*} d(T, T^*)^{-\frac{m-n}{2}}\right)(\frac{1}{\text{snr}})^{\frac{m-n}{2}} \end{aligned} \tag{38}$$

3.3.4 Bounding the Estimation Error

We have developed a probabilistic condition for $\hat{T} \in \mathcal{T}_1$. Now we need to bound the estimation error when $\hat{T} \in \mathcal{T}_1$. We have the following result.

Lemma 13. *Under Assumption 1, 2 and 3, if $\hat{T} \in \mathcal{T}_1$, then*

$$\frac{||\hat{x} - x^*||_2}{||\epsilon||_2} \leq \frac{1}{\sqrt{\hat{r}} - C\sqrt{n}(1+\delta_3)} \tag{39}$$

w.p. at least $1 - 2\exp(-n\delta_3^2)$, where $\hat{r} = \text{rank}(\hat{T})$.

Proof. Recall from Lemma 7 than when $\hat{\tau} \in \mathcal{T}_1$ then

$$||\hat{T}A(\hat{x} - x^*)||_2^2 = ||P_{\hat{T}A}\epsilon||_2^2 \tag{40}$$

Now let the thin-SVD of $\hat{T}$ be $U_{\hat{T}}V_{\hat{T}}^T$, where $U_{\hat{T}}, V_{\hat{T}} \in \mathbb{R}^{m\times\hat{r}}$. Now let $B = V_{\hat{T}}^T A \in \mathbb{R}^{\hat{r}\times n}$ a matrix with i.i.d. Gaussian entries. Then $\hat{T}A = U_{\hat{T}}B$. Now we have

$$||\hat{x} - x^*||_2^2 \leq \frac{||P_{\hat{T}A}\epsilon||_2^2}{\lambda_{\min}(A^T\hat{T}^T\hat{T}A)} \leq \frac{||\epsilon||_2^2}{\lambda_{\min}(B^TB)} \tag{41}$$

Therefore

$$||\hat{x} - x^*||_2 \leq \frac{||\epsilon||_2}{\sigma_n(B)} \tag{42}$$

Utilizing Theorem 4.6.1 in [20], we have $\sigma_n(B) \geq \sqrt{\hat{r}} - C\sqrt{n}(1+\delta_3)$ w.p. at least $1-2\exp(-n\delta_3^2)$, by which we finish the proof. □

3.3.5 Main Result and Asymptotic Analysis

Utilizing Proposition 2 and applying Lemma 13 with $\hat{r} = m$, we arrive at our main result which characterizes the relation of signal-to-noise ratio and the estimation error in the full rank case.

Theorem 4. *Let $A \in \mathbb{R}^{m\times n}$ be a i.i.d standard Gaussian matrix, and the conditions for unique recovery (in noiseless case) of $\mathcal{V}$ under $\mathcal{T}$ are satisfied. Suppose all T's in $\mathcal{T}$ are orthogonal. Then for any $\delta_1, \delta_2, \delta_3 \geq 0$, if*

$$\textit{snr} = \frac{||x^*||_2}{||\epsilon||_2} > \frac{4(m-1)(1+\delta_2)}{(m-n)\delta_1\sqrt{m}\cdot\min_{T\in\mathcal{T}\setminus T^*} d(T,T^*)} \tag{43}$$

then

$$\textit{err} = \frac{||\hat{x} - x^*||_2}{||\epsilon||_2} \leq \frac{1}{\sqrt{m} - C\sqrt{n}(1+\delta_3)} \tag{44}$$

w.p. at least

$$\begin{aligned} &1 - |\mathcal{T}|(Cm\delta_1^{1/2} + 2\exp(-Cm\delta_2^2)) - 2\exp(-n\delta_3^2) \\ &- 2^{2m-n-1}\left(\frac{1+\delta_2}{\delta_1\sqrt{m}}\right)^{\frac{m-n}{2}} \cdot \left(\sum_{T\in\mathcal{T}\setminus T^*} d(T,T^*)^{-\frac{m-n}{2}}\right)(\frac{1}{\textit{snr}})^{\frac{m-n}{2}} \end{aligned} \tag{45}$$

Let us analyze the bound here asymptotically. Note first that although the term 2^{2m-n-1} may seem devastating, it can in fact be easily rendered when m is bigger than some universal constants; for example when $m > 256$, $\sqrt{m}^{\frac{m-n}{2}} > 2^{2m-n-1}$, therefore we can now replace $\sqrt{m}$ by $C\sqrt{m}$ and ignore 2^{2m-n-1}. Now for the term involved with δ_2, for $|\mathcal{T}|\exp(-Cm\delta_2^2)$ to remain small, we need to have $\delta_2 = \Theta(\sqrt{\frac{\log|\mathcal{T}|}{m}})$; for $|\mathcal{T}|Cm\delta_1^{1/2}$ to remain small, we need to have $\frac{1}{\delta_1} = \Theta(m^2|\mathcal{T}|^2)$. Finally, δ_3 does not appear in the last term and thus relevant terms are negligible asymptotically. Therefore by plugging in what we have so far into both (43) and (45) we can see that the snr requirement for the probability to go towards 1 asymptotically is

$$\text{snr} = \Theta\left[m\cdot|\mathcal{T}|^2(\sqrt{m} + \sqrt{\log|\mathcal{T}|})\cdot\chi(T^*,\mathcal{T},m,n)\right] \tag{46}$$

where

$$\chi(T^*,\mathcal{T},m,n) = \max\left(\left(\sum_{T\in\mathcal{T}\setminus T^*} d(T,T^*)^{-\frac{m-n}{2}}\right)^{\frac{2}{m-n}}, \frac{m}{(m-n)\min_{T\in\mathcal{T}\setminus T^*} d(T,T^*)}\right) \tag{47}$$

We see that this result accords highly with our intuitions in following aspects:

- the snr requirement grows as m,$|\mathcal{T}|$ grows and n decreases.
- the snr requirement grows as $d(T,T^*)$'s decrease, i.e., the more "similar" the transformations are, the more strict the snr requirement is.

In linear regression without correspondence, $\mathcal{T}$ is the set of permutation matrices for which we denote by Π instead. Fix some Π^*. Since the number of Π's that achieve $d(\Pi^*, \Pi) = k$ is no more than m^k and the second term in (47) is obviously smaller than the first term, it can be seen that a upper bound for (47) is

$$\left(\sum_{2 \leq k \leq m} m^k (m/k)^{\frac{m-n}{2}} \right)^{\frac{2}{m-n}} \tag{48}$$

we now plot the snr requirement under this upper bound in Figure 1. It can be seen that the snr requirement grows as m/n grows, and when fixing m/n it also grows with m, n.

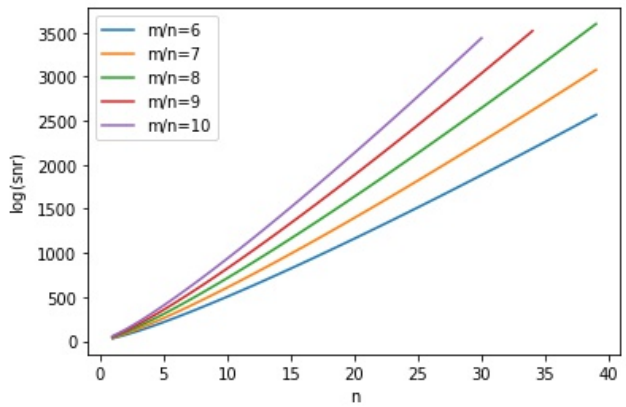

Figure 1: logarithm of snr requirement versus n and m/n

We leave detailed comparison of our bound and the bound given by analysis in instances of HS as future work.

3.3.6 Challenges from the Singular Case

Theorem 4 can only be applied to instances where the transformations in $\mathcal{T}$ are orthogonal, such as linear regression without correspondence/real phase retrieval. We expect that a similar result in the singular (i.e., non full-rank) case could be obtained. Recall from Equation (27) that we need an lower bound for $||T^*a||_2^2 - ||P_{Ta}T^*a||_2^2$. We assume that all T's have rank r. Now

$$\begin{aligned} ||T^*a||_2^2 - ||P_{Ta}T^*a||_2^2 &= ||T^*a||_2^2 - \frac{(a^T T^T T^* a)^2}{||Ta||_2^2} \\ &\geq ||T^*a||_2^2 - \frac{|a^T T^T T^* a| \cdot ||T^*a||_2}{||Ta||_2} \\ &= a^T T^{*T} T^* a - \frac{||T^*a||_2}{||Ta||_2} \cdot |a^T T^T T^* a| \end{aligned} \tag{49}$$

Now Utilizing Theorem 6.3.2 in [20], we have $\frac{1-\delta_1}{1+\delta_1} \leq \frac{||T^*a||_2}{||Ta||_2} \leq \frac{1+\delta_1}{1-\delta_1}$ w.p. at least $1 - 4\exp(-Cr\delta_1^2)$. Therefore for some positive value t, w.p. at least $1 - 4\exp(-Cr\delta_1^2)$,

$$\begin{aligned} &P(a^T T^{*T} T^* a - \frac{||T^*a||_2}{||Ta||_2} \cdot |a^T T^T T^* a| \leq t) \\ &\leq P(a^T T^{*T} T^* a - \frac{||T^*a||_2}{||Ta||_2} \cdot a^T T^T T^* a \leq t) + P(a^T T^{*T} T^* a + \frac{||T^*a||_2}{||Ta||_2} \cdot a^T T^T T^* a \leq t) \\ &\leq P(a^T T^{*T} T^* a - \frac{1+\delta_1}{1-\delta_1} \cdot a^T T^T T^* a \leq t) + P(a^T T^{*T} T^* a + \frac{1-\delta_1}{1+\delta_1} \cdot a^T T^T T^* a \leq t) \end{aligned} \tag{50}$$

where we have reached a situation similar in Equation (32). The big technical difficulty is that although the terms $\frac{1+\delta_1}{1-\delta_1}$ and $\frac{1-\delta_1}{1+\delta_1}$ can be afforded to approach 1, they can never exactly be 1 which

makes further analysis (in the way similar in (32)) difficult to perform (although this is not a problem for asymptotically analysis). We believe that this should be fixable, and expect that a bound similar in spirit to that in Theorem 4 which applies to the general case could be obtained. We leave this for future work.

4 Conclusion

We extended homomorphic sensing to the case where the input space is a union of linear spaces, which gives uniqueness guarantees for a class of sparse signal recovery problems. We also studied HS under noise, where we 1) developed deterministic conditions which guarantee small estimation error; 2) performed probabilistic analysis which characterizes the relation of the signal-to-noise ratio and the estimation error. Our current probabilistic analysis applies to cases where the transformations are orthogonal, but we expect that a similar result for transformations with 1/0 singular values could be obtained, which we put as future work.

References

[1] Abubakar Abid and James Zou. A stochastic expectation-maximization approach to shuffled linear regression. In *2018 56th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, pages 470–477. IEEE, 2018.

[2] Philip David, Daniel Dementhon, Ramani Duraiswami, and Hanan Samet. Softposit: Simultaneous pose and correspondence determination. *International Journal of Computer Vision*, 59(3):259–284, 2004.

[3] Joe Harris. *Algebraic geometry: a first course*. Springer-Verlag, 1992.

[4] Daniel J. Hsu, Kevin Shi, and Xiaorui Sun. Linear regression without correspondence. In *Advances in Neural Information Processing Systems*, pages 1531–1540, 2017.

[5] Jason M. Klusowski, Dana Yang, and William D. Brinda. Estimating the coefficients of a mixture of two linear regressions by expectation maximization. *IEEE Transactions on Information Theory*, 65(6):3515–3524, 2019.

[6] Partha Lahiri and Michael D. Larsen. Regression analysis with linked data. *Journal of the American statistical association*, 100(469):222–230, 2005.

[7] Xiaodong Li and Vladislav Voroninski. Sparse signal recovery from quadratic measurements via convex programming. *SIAM Journal on Mathematical Analysis*, 45(5):3019–3033, 2013.

[8] Wei Lian, Lei Zhang, and Ming-Hsuan Yang. An efficient globally optimal algorithm for asymmetric point matching. *IEEE Transactions on Pattern Analysis & Machine Intelligence*, pages 1–1, 2016.

[9] Fusheng Lv and Wenchang Sun. Real phase retrieval from unordered partial frame coefficients. *Advances in Computational Mathematics*, 2017.

[10] Manuel Marques, Marko Stosic, and Joa Costeira. Subspace matching: Unique solution to point matching with geometric constraints. In *IEEE International Conference on Computerp Vision*, 2009.

[11] Henrik Ohlsson and Yonina C. Eldar. On conditions for uniqueness in sparse phase retrieval. *IEEE International Conference on Acoustics Speech & Signal Processing*, 2013.

[12] Ashwin Pananjady, Martin J. Wainwright, and Thomas A. Courtade. Linear regression with shuffled data: Statistical and computational limits of permutation recovery. *IEEE Transactions on Information Theory*, 64(5):3286–3300, 2017.

[13] Aubrey B. Poore and Sabino Gadaleta. Some assignment problems arising from multiple target tracking. *Mathematical and Computer Modelling*, 43(9):1074–1091, 2006.

[14] Steven Roman. *Advanced linear algebra*. Springer, 2005.

[15] Xu Shi, Xiaoou Li, and Tianxi Cai. Spherical regression under mismatch corruption with application to automated knowledge translation. *Journal of the American Statistical Association*, (just-accepted):1–28, 2020.

[16] Martin Slawski and Emanuel Ben-David. Linear regression with sparsely permuted data. *Electronic Journal of Statistics*, 2017.

[17] Manolis C. Tsakiris and Liangzu Peng. Homomorphic sensing. In *International Conference on Machine Learning*, 2019.

[18] Manolis C. Tsakiris, Liangzu Peng, Aldo Conca, Laurent Kneip, Yuanming Shi, and Hayoung Choi. An algebraic-geometric approach to shuffled linear regression. *arXiv preprint arXiv:1810.05440*, 2018.

[19] Jayakrishnan Unnikrishnan, Saeid Haghighatshoar, and Martin Vetterli. Unlabeled sensing with random linear measurements. *IEEE Transactions on Information Theory*, PP(99), 2015.

[20] Roman Vershynin. *High-Dimensional Probability: An Introduction with Applications in Data Science*. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge University Press, 2018.

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